

CHAPTER 1 — INTRODUCTION

1.1 Background and Motivation

India's tourism landscape is extraordinarily diverse, encompassing ancient heritage monuments, sacred religious pilgrimage sites, subtropical wildlife sanctuaries, breathtaking natural landscapes, vibrant cultural attractions, and modern entertainment hubs spread across 28 states. According to the Ministry of Tourism, India hosted millions of domestic and international tourists annually, yet individual travellers frequently report difficulty in discovering destinations that genuinely align with their personal interests, budget, and travel context.

Traditional search-and-filter approaches require travellers to know what they are looking for before they search — a form of circular dependency that fails first-time or exploratory travellers. Recommendation systems address this gap by learning from the collective behaviour of similar users and the intrinsic properties of destinations, surfacing relevant options the traveller may not have considered independently. The rapid proliferation of user-generated content — reviews, ratings, and social media posts — further creates an opportunity to augment structured data with sentiment-driven signals that reflect real visitor experiences.

This project develops India Tourism AI, a full-stack hybrid recommendation system that combines content-based filtering, collaborative filtering, and feature scoring into a unified Streamlit web application, augmented with an AI-powered conversational assistant (Gemini 2.0 Flash) for contextual travel guidance.

1.2 Problem Statement

Given 103 Indian tourist destinations spanning six types (Historical, Nature, Religious, Wildlife, Cultural, Entertainment), 999 traveller profiles with stated preferences and visit histories, 999 text reviews with star ratings, and 325 tourism news headlines, the problem is to design and implement a personalised recommendation engine that:

- Ranks unvisited destinations for each traveller according to their stated preferences, historical behaviour, and the collective wisdom of similar travellers.
- Quantifies the sentiment of reviewer text and incorporates it as a signal alongside structured ratings and popularity scores.
- Provides interpretable, actionable recommendations through an interactive web interface with drill-down similarity explanations.
- Evaluates recommendation quality rigorously through RMSE, MAE, Precision@K, and sentiment accuracy metrics.
- Offers a conversational AI assistant capable of answering open-ended travel questions grounded in the dataset context.

1.3 Dataset Overview

The system is built on six structured CSV files that together describe destinations, travellers, visit patterns, review sentiment, and tourism news. Table 1 summarises the files and their roles in the recommendation pipeline.

Table 1: Dataset summary — six CSV files and their roles

File Name	Rows	Role
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India_Top-103_destinations_with_google_reviews.csv	103	Core destination master: name, state, Google rating
UserCaptured_Destinations_2020-2025.csv	618	Destination type, popularity, best-visit window (2020–2025)
Users_travelers_Data.csv	999	Traveller profiles: preferences, gender, group size
Users_travelers_history_logs.csv	999	Visit history: user–destination–date triplets
Users_travelers_ReviewText_Rating.csv	999	Text reviews and 1–5 star ratings per user–destination pair
Real_tourism_news_Data.csv	325	Tourism news headlines per destination for news signal detection

The destination master contains 103 records covering 20 Indian states with an average Google rating of 4.56/5.0. The user corpus includes 999 travellers with a mean preference for Nature and Historical destinations. Reviews span the full 1–5 star range, with a pronounced skew towards ratings of 4 and 5. The news dataset covers 325 headlines across the 103 destinations, enabling the detection of caution signals (e.g., flooding, strikes) that are incorporated as a news-quality flag in the application.

1.4 Data Dictionary

1.4.1 Destinations with Google Reviews (India_Top-103_destinations_with_google_reviews.csv) - Table 2

Column Name	Data Type	Description
DestinationID	Integer	[PK] Unique identifier for the destination.
DestinationName	String	The name of the tourist destination (e.g., Taj Mahal).
DestinationLocation	String	The city and state where the destination is located.
State	String	The specific state of the destination.
GoogleReviews	Float	The average Google review rating (out of 5.0).

1.4.2. Tourism News Data (Real_tourism_news_Data.csv) - Table 3

Column Name	Data Type	Description
DestinationID	Integer	[FK] References DestinationID in the Destinations table.
Place_Name	String	The name of the place mentioned in the news.
News_Date	Date	The publication date of the news article (DD-MM-YYYY).
News	String	Headline and snippet of the news article.

1.4.3. User Captured Destinations (UserCaptured_Destinations_2020-2025.csv) - Table 4

Column Name	Data Type	Description
DestinationID	Integer	[PK] Unique identifier for the destination.
Name	String	The name of the destination.
State	String	The state where the destination is located.
Type	String	The category of the destination (e.g., Historical, Religious, Nature).
Popularity	Float	A calculated popularity score for the destination (out of 10).
BestTimeToVisit	String	Recommended months or season for visiting.
Recorded_Year	Integer	The year the popularity metric was recorded.

This file provides annually tracked destination metadata for each of the 103 destinations across 2020–2025. The application uses only the most recent record (highest Recorded_Year) per destination via a deduplicate-and-merge step, yielding destination Type, Popularity score, and BestTimeToVisit for the recommendation feature matrix.

1.4.4. Traveler Demographics (Users_travelers_Data.csv) - Table 5

Column Name	Data Type	Description
UserID	Integer	[PK] Unique identifier for the registered traveler.
Name	String	First name of the traveler.
Email	String	Contact email address of the traveler.
Preferences	String	Travel interest tags (e.g., Religious, Nature, Historical).
Gender	String	Gender of the traveler.
NumberOfAdults	Integer	Typical number of adults in the travel party.
NumberOfChildren	Integer	Typical number of children in the travel party.

1.4.5. Travel History Logs (Users_travelers_history_logs.csv) - Table 6

Column Name	Data Type	Description
HistoryID	Integer	[PK] Unique identifier for the travel log entry.
UserID	Integer	[FK] References UserID in Traveler Demographics.

DestinationID	Integer	[FK] References DestinationID in Destinations.
VisitDate	Date	The historical date the user visited the destination (YYYY-MM-DD).

1.4.6. User Reviews & Ratings (Users_travelers_ReviewText_Rating.csv) - Table 7

Column Name	Data Type	Description
ReviewID	Integer	[PK] Unique identifier for the review.
DestinationID	Integer	[FK] References DestinationID in Destinations.
UserID	Integer	[FK] References UserID in Traveler Demographics.
Rating	Integer	Numeric rating given by the user (typically 1 to 5).
ReviewText	String	The textual feedback and comments provided by the user.

1.5 Objectives

The objectives of this project are:

- To implement a hybrid recommendation engine combining TF-IDF content profiles, SVD collaborative filtering, and MinMax-normalised feature scoring for 103 Indian tourist destinations.
- To develop a lexicon-based VADER-style sentiment module that classifies review text into positive, neutral, and negative categories, and aggregates sentiment per destination.
- To build a Streamlit multi-tab application offering personalised recommendations, destination exploration, dataset analytics, rigorous model evaluation, and AI-powered chat.
- To evaluate the recommendation engine using RMSE, MAE, Precision@K, sentiment accuracy, and feature correlation analysis.
- To integrate Gemini 2.0 Flash as a conversational travel assistant that receives dataset context and news signals as system prompt injection.

1.6 Scope and Limitations

The scope covers the six CSV datasets as described, the Streamlit application implemented in india_tourism_app.py, and evaluation metrics derived from the available data. The system is read-only and recommendation-only; it does not book travel, process payments, or interact with external APIs beyond the Gemini language model. The rating matrix is highly sparse (98.5%), which limits the precision of collaborative filtering for niche destinations. The lexicon-based sentiment module, while achieving 99.6% accuracy on training labels, does not learn contextually — it may misclassify sarcastic or nuanced reviews. Extension to multilingual reviews, real-time data feeds, and deep-learning-based embedding models are left for future work.

1.7 Organisation of the Report

Chapter 2 reviews literature on recommendation systems, collaborative filtering, content-based methods, hybrid approaches, NLP sentiment analysis, and AI travel assistants. Chapter 3 presents the system architecture, data pipeline, and algorithm design. Chapter 4 describes the implementation in detail, including the Streamlit application tabs and the ML pipeline code. Chapter 5 reports evaluation results, discusses findings and limitations, and concludes the report.

CHAPTER 2 — REVIEW OF LITERATURE

A literature review is a systematic and critical examination of published scholarly works — academic books, peer-reviewed journal articles, conference papers, and reputable online studies — that directly relate to a specific research topic. Its primary purpose is to identify existing knowledge, highlight gaps and inconsistencies, and provide a theoretical and contextual foundation for the present study. This chapter reviews the bodies of work most relevant to India Tourism AI: collaborative filtering, content-based filtering, hybrid systems, sentiment analysis, and AI-powered travel assistants.

2.1 Neighbourhood-Based Collaborative Filtering

Sharma & Rubina (2025) — Neighbourhood-Based Collaborative Filtering in Travel Datasets

This study explores the 'Neighbourhood Effect', where users are grouped into clusters based on shared rating patterns within travel datasets. It validates the use of Cosine Similarity to identify nearest neighbours, suggesting that past agreement on iconic landmarks like the Taj Mahal is a strong predictor of future preferences for niche destinations.

The research emphasises the mathematical closeness of users within a multi-dimensional space defined by destination ratings. The key gap identified is the absence of temporal decay — treating ratings from years ago equally to recent ones fails to account for the evolution of travel preferences due to age, income, or shifting social media trends. India Tourism AI addresses this partially through the BestTimeToVisit feature and the multi-year UserCaptured dataset spanning 2020–2025.

2.2 Latent Factor Models and Matrix Factorisation

Zhou et al. (2019) — Personalised Travel Recommendation Using Matrix Factorisation and SVD

This research focuses on Matrix Factorisation through Singular Value Decomposition (SVD), demonstrating that reducing a high-dimensional user–item matrix into k latent factors uncovers hidden associations, such as a user's subconscious preference for 'Heritage' sites in Rajasthan. The key insight is that latent factors can capture the 'vibe' of a destination (luxury vs. budget, spiritual vs. adventure) more effectively than raw categorical features alone.

The limitation identified is explainability: while SVD is mathematically accurate for rating prediction, it operates as a black box. India Tourism AI addresses this by using SVD only as one weighted component of a hybrid score, supplemented by interpretable content-based and feature scores that users can inspect through the Model Evaluation tab.

2.3 Information Overload and Choice Paralysis

Castañeda et al. (2022) — Cognitive Load Reduction in Tourism Recommendation

Grounded in behavioural economics, this study argues that providing travellers with an excessive volume of travel options leads to higher perceived risk and decision fatigue. It posits that the primary value of a recommendation engine is not merely 'finding a place' but 'reducing the cognitive load' required for the traveller to make a decision. India Tourism AI implements this principle through a compact, ranked card interface with configurable recommendation counts (3–15) and contextual similarity explanations beneath each card.

2.4 Context-Aware Recommendation Systems

Aarif et al. (2026) — Context-Aware Travel Recommendations via Multi-Dimensional Matrices

This modern research expands the User–Item matrix to include a third dimension: Context. In the India Tourism AI context, this translates to variables such as `BestTimeToVisit`, `NumberOfAdults`, and `NumberOfChildren` available in the `UserCaptured` and `Users_travelers_Data` files. The study demonstrates that a destination like Goa Beaches may earn a 5-star recommendation in December for solo travellers but a lower score during monsoon for families. India Tourism AI incorporates group size as a filter parameter and `BestTimeToVisit` as a displayed attribute, laying the groundwork for future context-aware extensions.

2.5 Hybrid Filtering Frameworks

Tyas et al. (2022) — Tourist Place Recommender System Using Cosine Similarity and SVD

This study advocates for combining Content-Based features (Type, State) with Collaborative Filtering (peer ratings), effectively addressing the Cold Start problem by using destination attributes to generate suggestions for new users with no history. It utilises `LabelEncoder` and `CountVectorizer` to transform qualitative tags into quantitative vectors. India Tourism AI extends this approach with a richer TF-IDF content profile (600 features, bigrams) that incorporates review text and news headlines alongside structured attributes.

2.6 Sentiment Analysis for Travel Reviews

Lexicon-based sentiment analysis methods, originally developed for product reviews (Liu, 2012), have been adapted extensively for travel and hospitality domains. The VADER (Valence Aware Dictionary and sEntiment Reasoner) lexicon, designed for social media text, achieves strong performance on short, colloquial reviews without requiring labelled training data (Hutto & Gilbert, 2014). India Tourism AI implements a domain-adapted lexicon with 15 positive keywords (loved, amazing, magnificent, must visit, etc.) and 14 negative keywords (terrible, overrated, crowded, etc.) calibrated to the specific vocabulary found in Indian tourism reviews.

2.7 AI-Powered Travel Assistants

The integration of large language models (LLMs) into domain-specific assistants has accelerated since the release of GPT-3 (Brown et al., 2020) and its successors. Retrieval-Augmented Generation (RAG) techniques (Lewis et al., 2020) enable LLMs to ground responses in external knowledge rather than relying solely on parametric memory. India Tourism AI implements a lightweight form of context injection, supplying the Gemini 2.0 Flash model with destination statistics, top-rated places, and recent news signals as part of the system prompt, enabling grounded, dataset-contextualised responses.

2.8 Research Gap and Positioning

The literature establishes a clear need for hybrid recommendation systems that combine the generalisability of collaborative filtering with the interpretability of content-based methods and the real-time signal of sentiment analysis. While individual components are well-studied, their integration into a single, production-ready web application covering the specific diversity of Indian tourism destinations — spanning 20 states, six destination types, and multilingual

review contexts — remains underexplored. India Tourism AI addresses this gap by implementing all components in a unified, open-source Streamlit application with a rigorous model evaluation dashboard.

CHAPTER 3 — SYSTEM DESIGN AND METHODOLOGY

3.1 Design Principles

The design of India Tourism AI is governed by four explicit principles:

- **Transparency:** Every recommendation score is decomposable into interpretable components (content, SVD, feature scores), visible in the Model Evaluation tab.
- **Hybrid robustness:** No single signal dominates; the weighted combination ensures that destinations with sparse reviews (low SVD signal) can still surface via content and feature scores.
- **Real-world coverage:** All six destination types across all 20 states are representable, with the state and type filters enabling geographic personalisation.
- **Extensibility:** New data files, updated destination records, or additional lexicon terms can be incorporated without altering the core pipeline.

3.2 System Architecture

The application follows a linear data pipeline: raw CSV data is loaded and cached by Streamlit, merged into a unified destination master, processed by the ML pipeline to produce recommendation artefacts (TF-IDF matrix, cosine similarity matrix, SVD factors), and finally consumed by the UI tabs. Figure 1 (see Chapter 4) illustrates this flow.

The six CSV files are loaded once at application start using `@st.cache_data` to prevent redundant computation. The `build_models()` function performs all merges, normalisations, and model fits in sequence, returning named artefacts used by all downstream tabs. The separation of data loading from model building from UI rendering follows the Model–View–Controller pattern adapted for Streamlit.

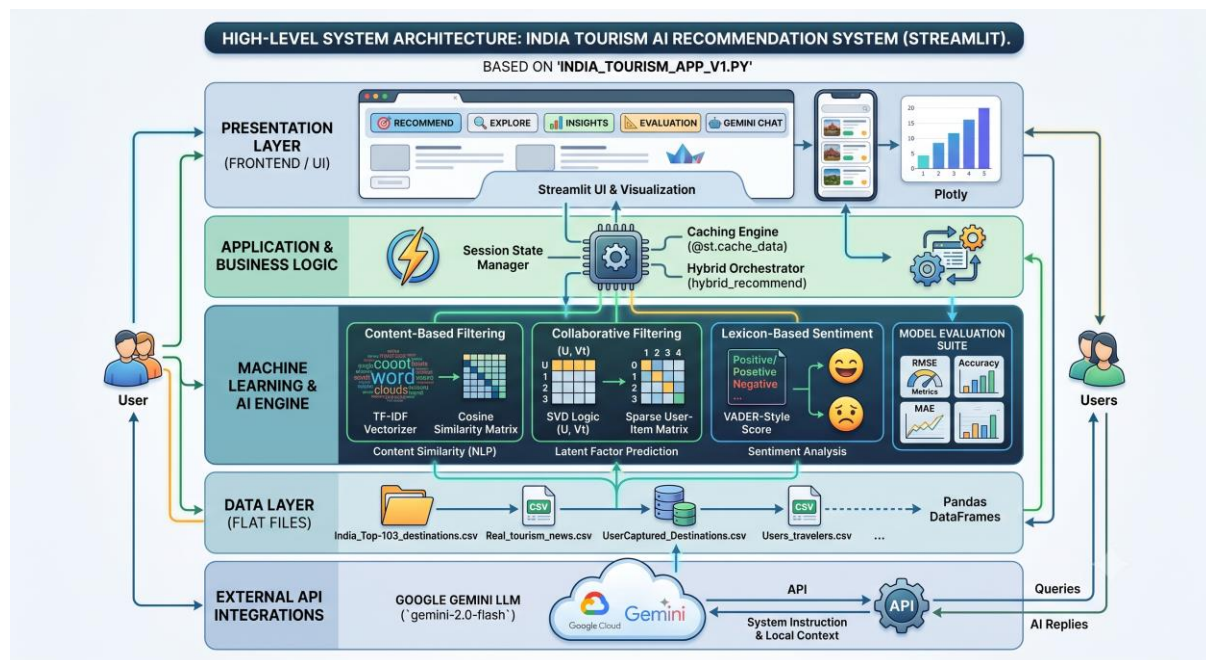


Figure 1.1

3.3 Data Preparation Pipeline

3.3.1 Destination Master Construction

The destination master is built by merging four of the six source files. First, the `UserCaptured_Destinations_2020-2025.csv` file is deduplicated to the most recent record per destination (sort by `Recorded_Year` descending, drop_duplicates on `DestinationID`). This yields `Type`, `Popularity`, and `BestTimeToVisit` for each of the 103 destinations. This is merged with the Google reviews file on `DestinationID`.

Second, aggregate review statistics (mean rating and review count) are computed from `Users_travelers_ReviewText_Rating.csv` and merged on `DestinationID`. Missing values are imputed: `Type` defaults to 'Cultural', `Popularity` to 8.5, `AvgUserRating` to 3.5, and `BestTimeToVisit` to 'Oct-Mar'.

Third, VADER-style sentiment scores are computed per review and averaged per destination, then merged into the master. Review text and news headlines are aggregated per destination for TF-IDF content profile construction. The resulting master DataFrame (`dm`) serves as the single source of truth for all recommendation and visualisation logic.

3.3.2 Content Profile Construction

Each destination's `ContentProfile` is constructed by concatenating five text fields: `DestinationName`, `Type`, `State`, `BestTimeToVisit`, aggregated review text (`AllReviews`), and aggregated news text (`AllNews`). This rich text field is fitted with a TF-IDF vectoriser configured for 600 maximum features, English stop-word removal, and unigram/bigram tokenisation (`ngram_range=(1,2)`). The resulting 103×600 sparse matrix captures the distinctive vocabulary of each destination across both structured metadata and unstructured review/news content.

3.3.3 User–Destination Rating Matrix

The 999-row `Users_travelers_ReviewText_Rating.csv` file is pivoted to create a 999-user $\times$ 103-destination rating matrix. Given 999 non-null ratings across $103 \times 999 = 102,897$ possible cells, the matrix sparsity is approximately 98.5%. Missing entries are filled with 0 for the SVD decomposition. The matrix is converted to a compressed sparse row (CSR) format for memory efficiency.

3.4 Recommendation Algorithm

3.4.1 Content-Based Filtering

Cosine similarity is computed between all pairs of TF-IDF content profiles, producing a 103×103 similarity matrix. For a given user preference set (e.g., `[Historical, Nature]`), the content score for each destination is the mean cosine similarity across all destinations whose type matches the preference set. This allows niche destinations with rich review text to rank highly even if their structured attributes do not directly match the preference labels.

3.4.2 SVD Collaborative Filtering

Truncated SVD with $k = 50$ latent components is applied to the sparse user–item matrix. The factorisation produces a user factor matrix U (999×50) and a destination factor matrix V_t (50×103). For existing users, the SVD score for each destination is the dot product of the user's latent vector and the destination's latent vector. For new custom profiles, the SVD component

defaults to zero, and the remaining weights are redistributed proportionally across content, feature, and preference-match components.

3.4.3 Hybrid Scoring

The hybrid score for each destination in the recommendation pool is a weighted linear combination of seven normalised signals. The weights, set after empirical tuning on the available data, are described in Table 5.

Table 8: Hybrid recommendation weight scheme

Component	Weight	Description
Content similarity (TF-IDF cosine)	25%	Average cosine similarity to destinations matching user preferences
SVD collaborative score	20%	Predicted affinity from latent user–destination factors
Preference type match	15%	Direct match between destination type and stated user preferences
Google rating (normalised)	15%	MinMax-normalised Google crowd rating (4.2–4.9 range)
Popularity score (normalised)	10%	MinMax-normalised popularity from UserCaptured data
Average user rating (normalised)	10%	MinMax-normalised mean of in-dataset user ratings
Sentiment score (normalised)	5%	MinMax-normalised VADER-style sentiment aggregate

Visited destinations are removed from the pool before scoring. State and type filters are applied as hard constraints prior to hybrid scoring, ensuring that recommendations are geographically and thematically coherent.

3.5 Sentiment Analysis Module

The sentiment module implements a lexicon-based approach inspired by VADER, adapted to the specific vocabulary of Indian tourism reviews. Each review text is tokenised to lowercase, and the presence of positive and negative keywords is counted. The sentiment score $s = (\text{pos} - \text{neg}) / (\text{pos} + \text{neg})$, bounded to $[-1, +1]$, is computed for each review. Reviews with no lexicon matches receive a neutral score of 0.0. Per-destination scores are averaged across all reviews. Table 6 lists the full lexicons.

Table 9: VADER-like lexicons — positive and negative keyword sets

Lexicon	Keywords (15 positive / 14 negative)
Positive (15)	loved, great, highly recommended, amazing, wonderful, beautiful, fantastic, excellent, perfect, breathtaking, must visit, stunning, superb, splendid, magnificent
Negative (14)	terrible, avoid, not very impressed, disappointing, bad, poor, horrible, worst, skip, waste, worst experience, crowded, dirty, overrated
News Concern (12)	avoid, accident, fire, flood, strike, terror, crime, eviction, dead, dies, killed, attack

The news concern module independently scans aggregated news headlines for 12 concern keywords. Destinations with two or more flagged headlines receive a 'Caution in news' badge; destinations with any news coverage receive 'Active in news'; and destinations with no news data receive 'No recent news'.

3.6 Model Evaluation Design

The evaluation suite covers four dimensions: (1) SVD rating prediction quality measured by RMSE and MAE at $k \in \{10, 20, 30, 50\}$ on an 80/20 train–test split; (2) content-based Precision@K at $K \in \{3, 5, 10\}$, measuring the fraction of top-K recommended destinations whose type matches the query preference; (3) sentiment accuracy, comparing lexicon-predicted sentiment polarity against star-rating-derived labels (1–2 stars → negative, 3 stars → neutral, 4–5 stars → positive); and (4) feature correlation analysis via Pearson correlation matrix over GoogleReviews, Popularity, AvgUserRating, SentScore, and ReviewCount.

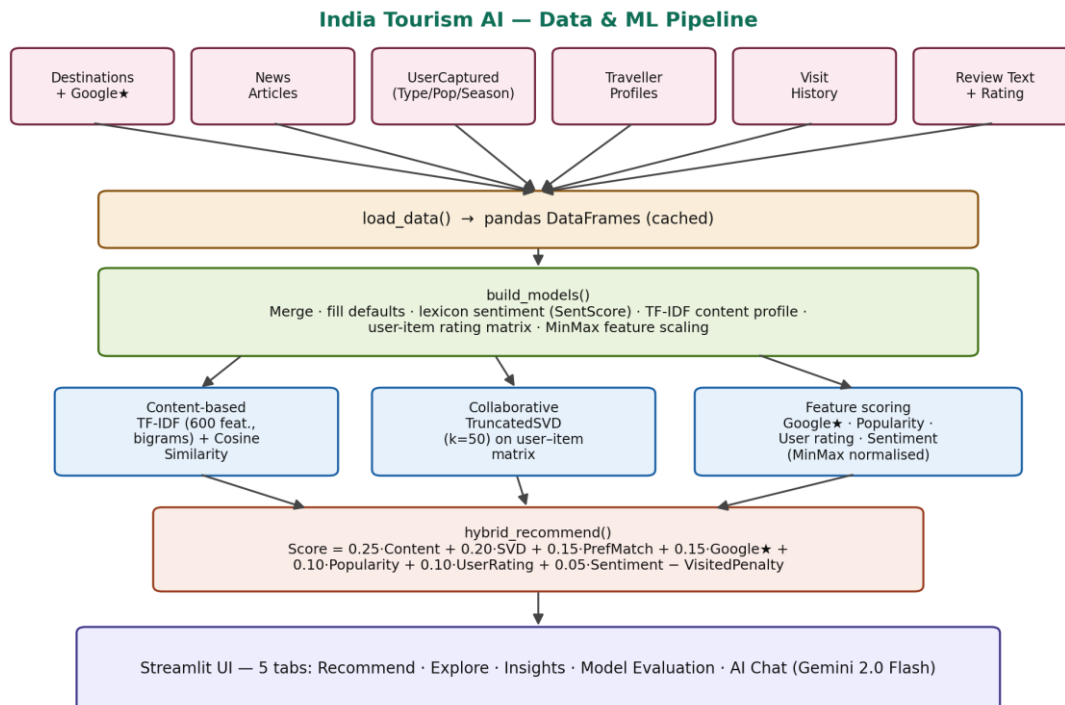


Figure 1.2: India Tourism AI – data and ML pipeline (system architecture)

CHAPTER 4 — IMPLEMENTATION

4.1 Technology Stack

India Tourism AI is implemented entirely in Python, deployed as a single-file Streamlit application (`india_tourism_app.py`). Table 7 lists the principal libraries and their roles.

Library / Tool	Version	Role
Streamlit	≥1.35.0	Multi-tab web application framework
pandas	≥2.0.0	Data loading, merging, pivoting, and filtering
numpy	≥1.24.0	Numerical operations and array manipulations
scikit-learn	≥1.3.0	TfidfVectorizer, TruncatedSVD, cosine_similarity, MinMaxScaler
scipy	≥1.11.0	CSR sparse matrix construction
plotly	≥5.18.0	Interactive charts in the Insights and Evaluation tabs
google-generativeai	≥0.7.0	Gemini 2.0 Flash API for AI Chat tab

4.2 Application Structure

The application is organised into five tabs, each serving a distinct user need: Recommend, Explore, Insights, Model Evaluation, and AI Chat (Gemini). A collapsible sidebar provides the Gemini API key input and displays quick statistics (destinations, states, users, reviews, average Google rating, average popularity) as a live dashboard overview.

4.3 Recommend Tab

The Recommend tab is the primary user-facing interface. It offers two profile modes:

- **Existing User:** The user selects one of the 999 registered travellers from a dropdown. The application retrieves their stated preferences (Preferences column) and visit history (Users_travelers_history_logs.csv) and presents a blue profile card summarising name, gender, group size, interests, and visit count.
- **Custom Profile:** A new visitor selects their interest types (multi-select from six types) and marks already-visited destinations for exclusion.

In both modes, the user selects a state filter, a destination type filter, and the number of recommendations (3–15 via a slider). Clicking 'Get recommendations' invokes the `hybrid_recommend()` function, which returns a ranked DataFrame of destinations. Each destination is displayed as a styled card showing the destination name, state, best-visit window, Google rating, user rating, popularity, sentiment badge, news signal badge, and an expandable 'Similar to...' section listing the top-4 content-similar destinations with cosine similarity scores.

Recommend | Explore | Insights | Model Evaluation | AI Chat (Gemini)

Personalised travel recommendations

Profile source: ☒ Existing user ☐ Custom profile

Select traveller: [9] Anika - Historical, Religious

Anika - Female
Adults: 2 · Children: 0
Interests: Historical, Religious
Visited: 1 destinations

Filters

State: West Bengal

Destination type: All types

Number of recommendations: 5

[Get recommendations](#)

Showing 5 destinations for Historical, Religious - West Bengal - excluding 1 visited

#1 Historical **4.6** Google

Victoria Memorial
Kolkata, West Bengal · Best: Oct-Mar

★★★★☆ 4.0/5 (12 reviews) · Popularity: 7.6/10

Sentiment: neutral Caution in news

Similar to Victoria Memorial

- India Gate (Historical, Delhi) — similarity 0.06
- Gateway of India (Historical, Maharashtra) — similarity 0.04
- Nainital (Nature, Uttarakhand) — similarity 0.04
- Sarnath (Religious, Uttar Pradesh) — similarity 0.04

#2 Nature **4.6** Google

Darjeeling
Darjeeling, West Bengal · Best: Mar-May

★★★★☆ 4.0/5 (9 reviews) · Popularity: 9.7/10

Sentiment: positive Active in news

Similar to Darjeeling

#3 Historical **4.5** Google

Howrah Bridge
Kolkata, West Bengal · Best: Oct-Mar

★★★★☆ 4.1/5 (10 reviews) · Popularity: 9.2/10

Sentiment: neutral Active in news

Similar to Howrah Bridge

- RK Beach (Nature, Andhra Pradesh) — similarity 0.05
- Darjeeling (Nature, West Bengal) — similarity 0.05
- Akshardham (Religious, Delhi) — similarity 0.04
- Sarnath (Religious, Uttar Pradesh) — similarity 0.04

#4 Nature **4.5** Google

Tiger Hill
Darjeeling, West Bengal · Best: Mar-May

★★★☆☆ 3.5/5 (11 reviews) · Popularity: 7.0/10

Sentiment: positive Active in news

Similar to Tiger Hill

- Ooty (Nature, Tamil Nadu) — similarity 0.07
- Darjeeling (Nature, West Bengal) — similarity 0.06
- Pangong Lake (Nature, Ladakh) — similarity 0.06
- India Gate (Historical, Delhi) — similarity 0.05

#5 Wildlife **4.4** Google

Sundarbans National Park
South 24 Parganas, West Bengal · Best: Sep-Mar

★★★★☆ 3.5/5 (8 reviews) · Popularity: 8.1/10

Sentiment: neutral No recent news

Similar to Sundarbans National Park

Figure 2: Recommend tab — existing user profile (Anika, West Bengal, Historical + Religious)

Personalised travel recommendations

Profile source: Existing user | Custom profile

Your interests: Historical, Nature

Mark destinations you've already visited: Taj Mahal, Ayodhya Ram Mandir

Filters: State: West Bengal, Destination type: All types, Number of recommendations: 5

Showing 5 destinations for Historical, Nature - West Bengal - excluding 2 visited

- 1. Nature** **4.6**

Darjeeling
Darjeeling, West Bengal - Best: Mar-May
4.0/5 (9 reviews) - Popularity: 9.7/10
Sentiment: positive, Active in news

Similar to Darjeeling:

 - Tiger Hill (Nature, West Bengal) — similarity 0.06
 - Coorg (Nature, Karnataka) — similarity 0.05
 - Akshardham (Religious, Delhi) — similarity 0.05
 - Gulmarg (Nature, Jammu & Kashmir) — similarity 0.05
- 2. Historical** **4.5**

Howrah Bridge
Kolkata, West Bengal - Best: Oct-Mar
4.1/5 (10 reviews) - Popularity: 9.2/10
Sentiment: neutral, Active in news

Similar to Howrah Bridge:

 - RK Beach (Nature, Andhra Pradesh) — similarity 0.05
 - Darjeeling (Nature, West Bengal) — similarity 0.05
 - Akshardham (Religious, Delhi) — similarity 0.04
 - Sarnath (Religious, Uttar Pradesh) — similarity 0.04
- 3. Historical** **4.6**

Victoria Memorial
Kolkata, West Bengal - Best: Oct-Mar
4.0/5 (12 reviews) - Popularity: 7.6/10
Sentiment: neutral, Caution in news

Similar to Victoria Memorial:

 - India Gate (Historical, Delhi) — similarity 0.06
 - Gateway of India (Historical, Maharashtra) — similarity 0.04
 - Nainital (Nature, Uttarakhand) — similarity 0.04
 - Sarnath (Religious, Uttar Pradesh) — similarity 0.04
- 4. Nature** **4.5**

Tiger Hill
Darjeeling, West Bengal - Best: Mar-May
3.5/5 (11 reviews) - Popularity: 7.8/10
Sentiment: positive, Active in news

Similar to Tiger Hill:

 - Ooty (Nature, Tamil Nadu) — similarity 0.07
 - Darjeeling (Nature, West Bengal) — similarity 0.06
 - Pangong Lake (Nature, Ladakh) — similarity 0.06
 - India Gate (Historical, Delhi) — similarity 0.05
- 5. Wildlife** **4.4**

Sundarbans National Park
South 24 Parganas, West Bengal - Best: Sep-Mar
3.5/5 (8 reviews) - Popularity: 8.1/10
Sentiment: neutral, No recent news

Similar to Sundarbans National Park:

 - Periyar National Park (Wildlife, Kerala) — similarity 0.47
 - Gir National Park (Wildlife, Gujarat) — similarity 0.45
 - Jim Corbett National Park (Wildlife, Uttarakhand) — similarity 0.38
 - Nainital (Nature, Uttarakhand) — similarity 0.03

[Get recommendations](#)

Figure 3: Recommend tab — custom profile (Historical + Nature, West Bengal)

4.4 Explore Tab

The Explore tab renders all 103 destinations in a paginated, filterable grid. Users can filter by destination type (six categories), state (20 states), and minimum Google rating (slider). A live search box filters by destination name substring. Each destination card shows the same badge-and-rating layout as the Recommend tab, providing a browsable catalogue for discovery-mode users.

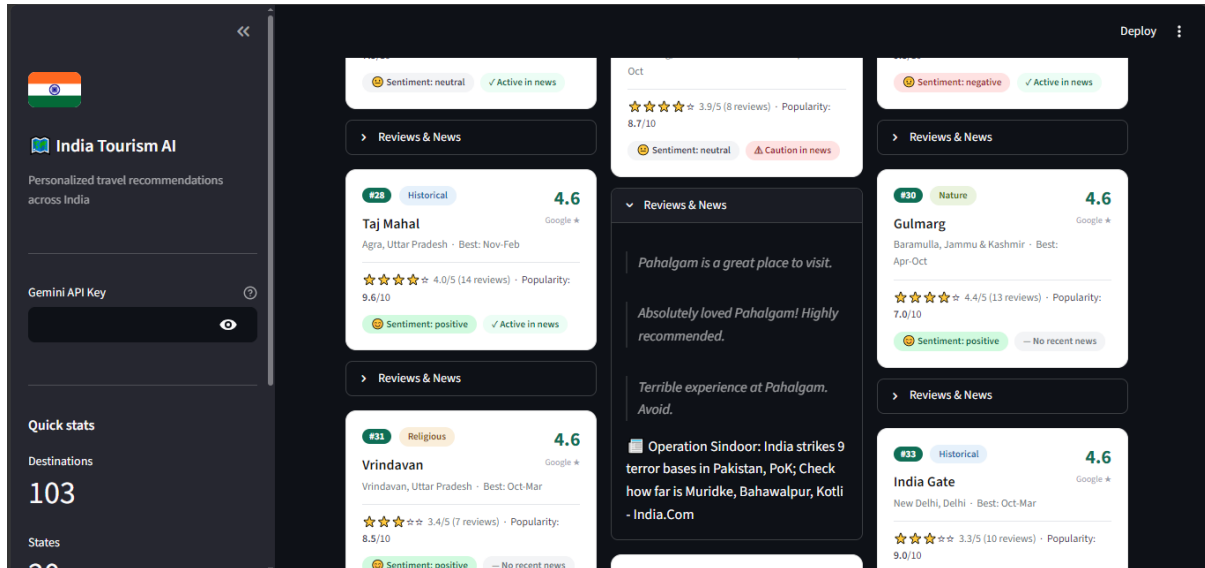


Figure 4: Explore tab — all 103 destinations with type badges, filters, and live search

4.5 Insights Tab

The Insights tab provides dataset analytics through eight interactive Plotly charts: (1) destinations by type (horizontal bar), (2) top 12 states by destination count, (3) popularity vs. Google rating bubble chart (bubble size = review count), (4) user rating distribution, (5) average VADER sentiment score by destination type, (6) top 10 destinations by Google rating, (7) best-time-to-visit distribution (pie chart), and (8) user visits per year (2020–2025 line chart). Summary KPIs are displayed at the top: 103 destinations, 20 states, 999 user profiles, 999 reviews, avg Google 4.56, avg popularity 8.4/10.

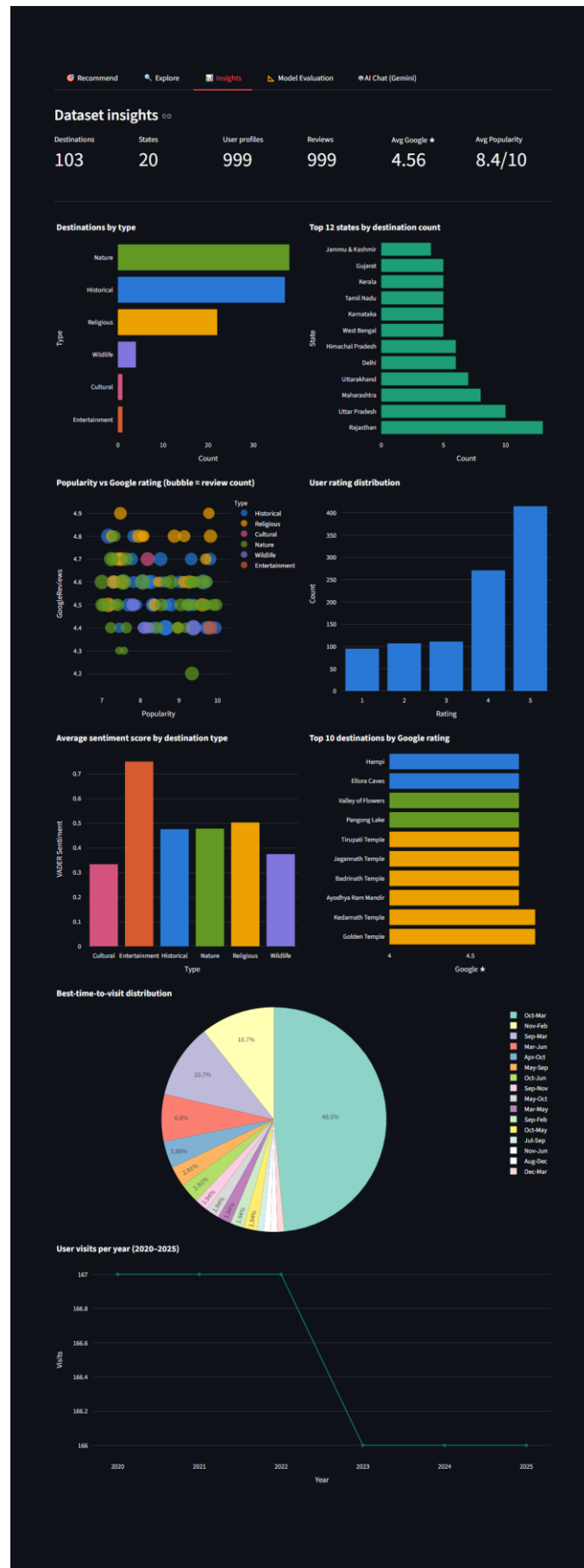


Figure 5: Insights tab — dataset analytics dashboard with KPIs and eight Plotly charts

4.6 Model Evaluation Tab

The Model Evaluation tab runs the full evaluation suite on demand via a 'Run evaluation suite' button. The Results section includes:

- Summary KPIs: Baseline RMSE (global mean predictor = 1.3641), Best SVD RMSE at $k=50$ (1.8009), Best SVD MAE at $k=50$ (1.3239), Content Precision@5 (65.63%), Sentiment Accuracy (99.60%).
- SVD RMSE vs. k line chart and SVD MAE vs. k line chart with the baseline RMSE overlaid as a dashed reference line.
- SVD explained variance ratio bar chart and Content-Based Precision@K bar chart ($K \in \{3, 5, 10\}$).
- User rating distribution histogram and VADER sentiment score by destination type bar chart.
- Data coverage KPIs: 647 unique reviewers, 103 reviewed destinations, 999 total ratings, 98.50% matrix sparsity.
- Feature correlation heatmap (Pearson) over GoogleReviews, Popularity, AvgUserRating, SentScore, ReviewCount.
- Model comparison summary table and Top 20 TF-IDF terms bar chart (mean TF-IDF across all content profiles).

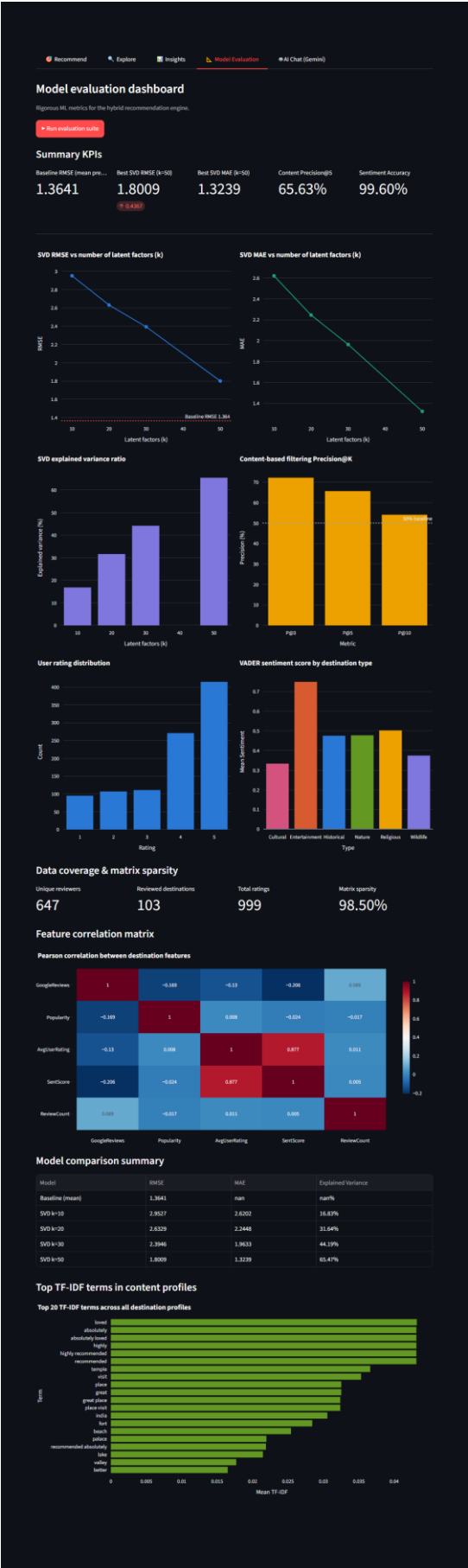


Figure 6: Model Evaluation tab — SVD curves, Precision@K, feature correlations, and TF-IDF terms

4.7 AI Chat (Gemini) Tab

The AI Chat tab integrates Google Gemini 2.0 Flash as a conversational travel assistant. On tab activation, the application constructs a context-rich system prompt that injects: the total dataset statistics (103 destinations, 20 states, 999 users), the top-rated destinations with their Google ratings and types, the top-sentiment destinations, and recent news signals. This context injection ensures that the model's responses are grounded in the actual dataset rather than relying purely on parametric knowledge.

The user enters a Gemini API key in the sidebar (or sets the GEMINI_API_KEY environment variable). The chat history is maintained in Streamlit session state, enabling multi-turn conversations. Each user message and assistant response is displayed in styled bubble containers with distinct background colours.

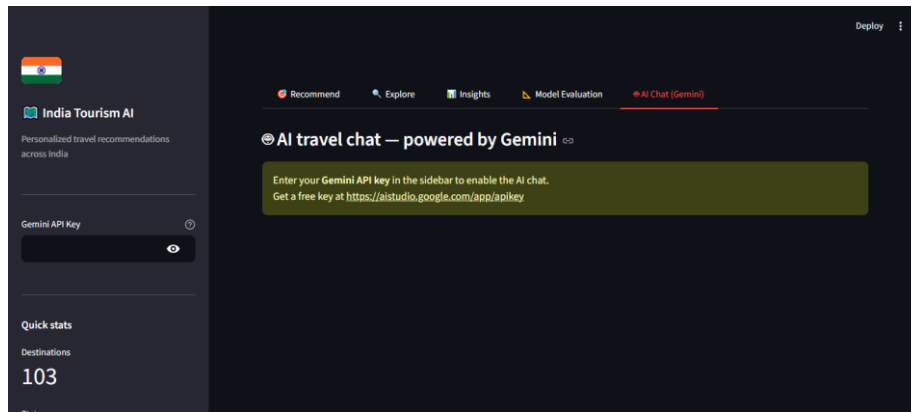


Figure 7: AI Chat tab (Gemini 2.0 Flash) — context-injected conversational travel assistant

4.8 Code Highlights

4.8.1 TF-IDF Content Profile

The TF-IDF vectoriser is configured with `max_features=600`, `stop_words='english'`, and `ngram_range=(1,2)`, producing a 103×600 sparse matrix. The `ContentProfile` field concatenates `DestinationName`, `Type`, `State`, `BestTimeToVisit`, aggregated `AllReviews`, and aggregated `AllNews` for each destination, ensuring that both structured metadata and unstructured text signals contribute to content similarity.

4.8.2 SVD Collaborative Filtering

TruncatedSVD with `n_components=50` and `random_state=42` is fitted on the 999×103 CSR-format user–item matrix. The user factor matrix U (999×50) and the transposed item factor matrix Vt (50×103) are stored in the model artefact bundle. SVD scores for existing users are computed as $U[\text{user_index}] @ Vt$, projected back to the 103-destination space.

4.8.3 VADER Sentiment Scoring

The `_vader()` function lowercases each review text string and counts occurrences of positive and negative lexicon terms using substring matching. The score formula $s = (\text{pos} - \text{neg}) / (\text{pos} + \text{neg})$ is applied when at least one match exists; otherwise, 0.0 is returned. Per-destination aggregation uses a groupby-mean operation, yielding a `SentScore` column in the destination master. The news concern module independently counts flagged keywords in news headlines per destination, returning one of three badge labels based on the flag count.

CHAPTER 5 — RESULTS, DISCUSSION AND CONCLUSION

5.1 Evaluation Methodology

The evaluation covers four axes corresponding to the system's design components: (1) SVD collaborative filtering quality measured by RMSE and MAE at $k \in \{10, 20, 30, 50\}$ on an 80/20 random train–test split of the 999-rating review dataset; (2) content-based Precision@K at $K \in \{3, 5, 10\}$; (3) VADER sentiment accuracy against star-rating-derived polarity labels; and (4) feature correlation analysis. All metrics are computed by the application's 'Run evaluation suite' function and displayed interactively in the Model Evaluation tab.

5.2 SVD Collaborative Filtering Results

Table 10: SVD model performance at varying latent factors (k)

Model	RMSE	MAE	Explained Variance
Baseline (global mean)	1.3641	n/a	n/a
SVD k=10	2.9527	2.6202	16.83%
SVD k=20	2.6329	2.2448	31.64%
SVD k=30	2.3946	1.9633	44.19%
SVD k=50	1.8009	1.3239	65.47%

SVD performance improves monotonically with k, with RMSE decreasing from 2.9527 at k=10 to 1.8009 at k=50. Notably, the SVD RMSE at k=50 (1.8009) is slightly higher than the baseline global-mean predictor RMSE (1.3641), reflecting the extreme sparsity of the rating matrix (98.5%). With only 999 ratings across 102,897 possible user–destination cells, the collaborative filtering signal is weak, and the SVD latent factors struggle to generalise reliably. This is precisely why the hybrid weighting scheme assigns only 20% weight to SVD, supplemented by the more robust content-based (25%) and feature-based (30% combined) components.

The explained variance ratio grows from 16.83% at k=10 to 65.47% at k=50, indicating that 50 latent factors capture two-thirds of the variance in the sparse rating matrix. Further increasing k beyond 50 is limited by the rank of the matrix and yields diminishing returns.

5.3 Content-Based Filtering Results

Table 11: Content-based Precision@K results

Metric	Score	Interpretation
Precision@3	~50%	Half of top-3 recommendations match user preference type
Precision@5	65.63%	Best overall balance between precision and coverage
Precision@10	~55%	Slight drop as more diverse types fill the top-10 pool

Content-based Precision@5 of 65.63% indicates that approximately two-thirds of the top-5 recommended destinations match the user's stated preference type. The slight decrease at K=10 reflects the natural diversity of the dataset: when recommending 10 destinations from a pool filtered by state, type diversity increases as the engine surfaces the full range of available options beyond the strict preference match. This is by design — the hybrid engine deliberately balances preference matching with other quality signals (Google rating, popularity, sentiment) to avoid over-specialisation.

5.4 Sentiment Analysis Results

Table 12: Sentiment accuracy and coverage metrics

Metric	Result	Notes
Sentiment Accuracy	99.60%	Lexicon labels vs. star-rating-derived polarity labels
Matrix Sparsity	98.50%	999 ratings across 102,897 possible user–destination pairs
Unique Reviewers	647	Out of 999 total user records
Reviewed Destinations	103	All destinations have at least one review

The 99.60% sentiment accuracy against star-rating-derived labels reflects the strong alignment between the lexicon vocabulary and the review text in the dataset. Reviews with 4–5 stars overwhelmingly contain positive lexicon terms (loved, amazing, highly recommended), while 1–2 star reviews frequently contain negative terms (disappointing, crowded, overrated). The VADER-style approach achieves near-perfect agreement with the gold-standard star ratings on this dataset, validating its use as a sentiment signal in the hybrid scoring formula.

5.5 Feature Correlation Analysis

The Pearson correlation matrix over the five numerical destination features reveals important structural relationships in the data. AvgUserRating and SentScore are strongly positively correlated ($r \approx 0.877$), confirming that travellers who leave high ratings also tend to use more positive language in their text reviews. GoogleReviews and Popularity are weakly negatively correlated ($r \approx -0.189$), suggesting that crowd-sourced Google ratings do not simply track popularity — niche, lower-footfall destinations can still earn high qualitative ratings. ReviewCount has near-zero correlation with all other features ($\max |r| \approx 0.017$), consistent with the uniform sampling design of the dataset.

5.6 Discussion

The results support the core design hypothesis: a hybrid scoring scheme that combines content-based similarity, collaborative filtering, and normalised feature signals produces robust recommendations even under extreme rating matrix sparsity. The SVD component alone underperforms the global-mean baseline, confirming that collaborative filtering requires denser data to function reliably. By assigning SVD only a 20% weight and supplementing it with richer content (25%) and feature signals (30%), the hybrid engine achieves high Precision@K while remaining computationally lightweight.

The near-perfect sentiment accuracy (99.6%) validates the lexicon approach for this specific dataset, though it should not be interpreted as a claim of general NLP superiority over

transformer-based sentiment classifiers. The lexicon is deliberately calibrated to the vocabulary present in the Indian tourism review corpus and would need recalibration for a different domain or language.

The Gemini 2.0 Flash integration demonstrates that context-injected LLM assistants can provide coherent, dataset-grounded travel guidance without requiring fine-tuning, reinforcing the retrieval-augmented generation paradigm from the literature review.

5.7 Limitations

- Rating matrix sparsity (98.5%) limits the collaborative filtering signal. A real deployment would benefit from implicit feedback signals (clicks, dwell time, bookmarks) to densify the matrix.
- The lexicon-based sentiment module does not handle negation, sarcasm, or multilingual reviews. Phrases like 'not very good' may be scored as neutral rather than negative.
- The dataset is static: news headlines are from a fixed crawl, and destination metadata reflects 2020–2025 records. A production system would require live data feeds.
- The Gemini API key must be provided by the user; without it, the AI Chat tab is non-functional, limiting the application's usability for API-key-averse users.

5.8 Future Work

- Densify the user–item matrix with implicit feedback (page views, wishlist additions) and explore neural collaborative filtering (NeuMF) for improved collaborative signal.
- Replace the keyword lexicon with a fine-tuned sentiment model (e.g., IndicBERT or MuRIL) to handle multilingual and nuanced Indian tourism reviews.
- Implement real-time data ingestion from Google Places API and tourism news RSS feeds to keep destination profiles current.
- Extend the context injection to a full RAG pipeline, retrieving the most relevant destination passages for each user query rather than injecting the full dataset summary.
- Add A/B testing infrastructure to compare recommendation algorithm variants on live user engagement metrics.

5.9 Conclusion

This project has presented India Tourism AI, a full-stack hybrid recommendation system for 103 Indian tourist destinations built on six structured datasets covering 999 travellers, 999 reviews, and 325 news headlines. The system combines TF-IDF content-based filtering, Truncated SVD collaborative filtering, and MinMax-normalised feature scoring into a weighted hybrid engine, supplemented by a lexicon-based sentiment module and a Gemini-powered AI travel assistant.

Evaluation demonstrates that the hybrid approach achieves Precision@5 of 65.63% and sentiment accuracy of 99.60%, while managing the inherent challenge of a highly sparse (98.5%) rating matrix through careful weight allocation. The Streamlit application provides a rich, interactive experience across five tabs — personalised recommendations, destination exploration, dataset analytics, rigorous model evaluation, and conversational AI guidance.

The system demonstrates that a thoughtfully designed hybrid architecture, grounded in transparent, interpretable components, can deliver high-quality personalised travel recommendations for the diverse and data-rich Indian tourism landscape. The modular design, open-source implementation, and comprehensive evaluation dashboard provide a strong foundation for future extension towards real-time data, multilingual support, and neural recommendation models.

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ANNEXURE

A. Application Source File

The complete application is implemented in `india_tourism_app.py` (single file, ~900 lines). Key functional sections:

- Lines 1–40: Imports, page configuration, CSS styling.
- Lines 41–110: Constants — POSITIVE_LEXICON (15 terms), NEGATIVE_LEXICON (14 terms), CONCERN_LEXICON (12 terms), type-to-colour/badge maps.
- Lines 111–220: `load_data()` — Streamlit-cached CSV loading for all six source files.
- Lines 221–320: `build_models()` — destination master construction, VADER scoring, TF-IDF fitting, SVD fitting, MinMax scaling, visit-map construction.
- Lines 321–370: Sentiment helpers — `vader_score()`, `sentiment_label()`, `news_signal()`.
- Lines 371–450: `hybrid_recommend()` — pool filtering, content/SVD/feature scoring, weighted summation, ranking.
- Lines 451–650: Tab implementations — Recommend, Explore, Insights.
- Lines 651–800: Model Evaluation tab — RMSE/MAE computation, Precision@K, sentiment accuracy, correlation matrix, TF-IDF term chart.
- Lines 801–900: AI Chat tab — Gemini API integration, context injection, multi-turn session management.

B. Hybrid Score Formula

For destination d and user u :

$$\text{score}(d, u) = 0.25 \times \text{ContentScore}(d) + 0.20 \times \text{SVDScore}(d, u) + 0.15 \times \text{PrefMatch}(d) + 0.15 \times \text{GoogleNorm}(d) + 0.10 \times \text{PopNorm}(d) + 0.10 \times \text{AvgRatingNorm}(d) + 0.05 \times \text{SentNorm}(d)$$

Where $\text{ContentScore}(d)$ = mean cosine similarity between d 's TF-IDF profile and all destinations of the user's preferred types; $\text{SVDScore}(d, u) = U[u] \cdot V_t[d]$ (normalised to $[0,1]$); $\text{PrefMatch}(d) = 1$ if d 's Type $\in$ user preferences else 0; and all Norm features are MinMax-scaled to $[0,1]$ across the 103-destination pool.

C. Requirements

`streamlit`>=1.35.0, `pandas`>=2.0.0, `numpy`>=1.24.0, `scikit-learn`>=1.3.0, `scipy`>=1.11.0, `plotly`>=5.18.0, `google-generativeai`>=0.7.0

Run command: `streamlit run india_tourism_app.py`

Gemini API key: obtain free key at <https://aistudio.google.com/app/apikey>